A Minor Project Synopsis

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SYNOPSIS REPORT

Minor Project

Explainable Academic Risk Prediction Using XGBoost and SHAP

Student Name	Yadav Hitesh Shravan, Mayank Jaiswal, Ritesh Raj, Shubham Thakur
Project Title	**Explainable Academic Risk Prediction**
Dataset	Students Performance Dataset (Kaggle — Rabie El Kharoua)
Proposed Algorithm	XGBoost with Hyperparameter Tuning
Explainability	SHAP (SHapley Additive exPlanations)
Baseline Algorithms	Logistic Regression, Random Forest
Deployment	Streamlit Web Application
Date	March 27, 2026

Abstract

Academic risk prediction has been widely studied using machine learning, but a critical gap persists: models predict whether a student is at risk but fail to explain why. This 'black box' problem limits educator trust and prevents effective intervention. Furthermore, no existing study applies Indian grading standards — where GPA below 2.0 represents the 50% passing threshold — to define academic risk. This project addresses both gaps by building an explainable AI system using XGBoost with SHAP that achieves 93.32% accuracy and 0.94 recall for at-risk students, while providing individual per-student explanations. Three algorithms are compared — Logistic Regression, Random Forest, and XGBoost — and the system is deployed as a Streamlit web application accessible to non-technical educators.

Section 1 | Introduction

Education is the foundation of social and economic development. Identifying students who are at risk of academic failure at an early stage is one of the most important challenges faced by educational institutions today. When students fall behind without anyone noticing, the consequences — poor grades, loss of confidence, and eventual dropout — can be avoided if the right intervention is made in time.

In India, a GPA of 2.0 on a 4.0 scale corresponds to approximately 50%, which is the standard passing threshold across most educational boards and universities. A student with a GPA below this threshold is considered academically at risk and requires immediate attention from teachers, counselors, and family.

Machine learning has emerged as a powerful tool for predicting such risk. By analyzing patterns in student data — including study habits, attendance, family background, and participation in activities — ML models can identify at-risk students before their grades deteriorate to a critical level.

However, most existing ML-based prediction systems suffer from a fundamental problem: they are black boxes. A teacher is told 'this student is at risk' but given no explanation as to why. Without understanding the reason, a teacher cannot take targeted action. Is it the absences? The lack of study time? Low parental support? The prediction alone is not enough.

This project addresses this critical gap by building an Explainable Academic Risk Prediction system. Using XGBoost — a high-performance gradient boosting algorithm — combined with SHAP (SHapley Additive exPlanations), the system not only predicts academic risk with 93.32% accuracy but also shows exactly which factors are driving the prediction for each individual student. The entire system is deployed as a browser-based web application using Streamlit, making it immediately usable by teachers and counselors without any programming knowledge.

1.1 Background

The use of machine learning in education — known as Educational Data Mining (EDM) — has grown significantly over the past decade. Researchers have applied algorithms like Decision Trees, Support Vector Machines, Random Forests, and Neural Networks to predict student performance. While these studies demonstrate that ML can effectively identify at-risk students, they consistently stop at reporting accuracy metrics without providing any practical tool for educators.

The emergence of explainable AI (XAI) tools like SHAP has opened a new opportunity to bridge this gap. SHAP provides mathematically grounded explanations for any ML model's predictions, making it possible to tell not just what the prediction is, but why it was made.

1.2 Motivation

The motivation for this project comes from a simple observation: a prediction that cannot be explained cannot be trusted, and a tool that cannot be used by teachers has no real-world value. Existing research has built accurate models but left them confined to academic papers.

This project is motivated by the desire to build something that actually reaches the teachers and counselors who need it.

1.3 Project Contribution

- First explainable academic risk prediction system using XGBoost + SHAP with Indian grading standards
- Individual per-student SHAP explanations deployed in a real-time Streamlit web application
- Rigorous comparison of three ML algorithms proving XGBoost's recall advantage for at-risk detection
- Practical tool usable by non-technical educators in any browser

Section 2 | Problem Statement

The problem this project addresses can be broken into three dimensions:

2.1 The Black Box Problem

Existing machine learning models for academic risk prediction are black boxes. They produce a binary output — At Risk or Not At Risk — but provide no explanation for their decision. Teachers are expected to act on predictions they cannot verify or understand. This lack of transparency undermines trust in the system and prevents effective use in real educational settings.

> **Gap 1: No existing academic risk prediction system provides per-student explanations that educators can understand and act upon.**

2.2 The Deployment Gap

A thorough review of existing literature reveals that even when accurate models are built, they are never deployed in a form usable by teachers or counselors. Research findings remain confined to papers and notebooks. The gap between model development and practical educational use has not been addressed by any existing study reviewed.

> **Gap 2: No existing study deploys an academic risk prediction system with explainability in a real-time web application for non-technical educators.**

2.3 The Indian Grading Standards Gap

The vast majority of research on academic risk prediction uses datasets and grading systems from Western countries — particularly Portugal, the United States, and Europe. No existing study reviewed applies Indian grading standards, where a GPA below 2.0 on a 4.0 scale (equivalent to 50%) defines the passing threshold and therefore the risk boundary.

> **Gap 3: No existing academic risk prediction system is designed for Indian educational standards using the GPA < 2.0 risk threshold.**

2.4 Problem Statement Summary

> *Build an explainable academic risk prediction system using XGBoost and SHAP that: (1) predicts binary academic risk using Indian grading standards, (2) provides individual per-student SHAP explanations for every prediction, and (3) deploys both the prediction and explanation in a real-time web interface accessible to non-technical educators.*

Section 3 | Literature Review

A systematic review of five recent research papers (2025-2026) was conducted to understand the current state of academic risk prediction and identify the specific gaps this project addresses.

Paper 1 — XGBoost + SHAP for Student Academic Performance

Published In	Journal of Artificial Intelligence and Engineering Applications, 2026
Algorithm Used	XGBoost with SHAP interpretability
Dataset Size	1,000 entries, 16 variables
Focus	General academic achievement prediction based on learning habits
Gap Identified	No binary risk classification, no Indian grading, no deployable interface for educators

Paper 2 — Interpretable ML for Academic Performance: SHAP Analysis

Published In	Innovations in Education and Teaching International, 2025
Algorithms Used	Random Forest, XGBoost, SVM, Neural Network
Focus	Predicting final exam scores of vocational students
Gap Identified	Predicted continuous scores, not binary risk. No deployable tool built.

Paper 3 — ML Models for Academic Performance: Interpretability

Published In	Frontiers in Education, August 2025
Algorithms Used	Random Forest, XGBoost with SHAP
Focus	Student performance prediction with interpretability analysis
Gap Identified	SHAP explanations produced but not embedded in any interactive real-time system

Paper 4 — XGBoost Early Warning System for Student Dropout

Published In	MDPI Applied Sciences, August 2025
Algorithms Used	XGBoost, Logistic Regression, Random Forest, LightGBM
Dataset Size	~40,000 students using LMS and enrollment data
Focus	Dropout prediction from day one using institutional data
Gap Identified	Dropout focus, not GPA-based risk. LMS data unavailable to most teachers. No Indian grading.

Paper 5 — Hybrid XGBoost + SHAP for School Dropout Prediction

Published In	PLOS One, September 2025
Algorithms Used	XGBoost, Random Forest with SHAP and LIME
Accuracy	XGBoost: 94.4%
Focus	School dropout prediction in Bangladesh — South Asian context
Gap Identified	No deployable interface. SHAP used for policy analysis only, not educator access. No Indian grading threshold.

3.6 Research Gap Summary

> **All five reviewed papers share a common gap: XGBoost and SHAP are used for prediction and analysis, but never deployed in a real-time interface where educators can access individual student explanations. No paper applies Indian grading standards. No paper builds a tool usable by non-technical teachers. This project fills all three gaps simultaneously.**

Section 4 | Objectives

The following objectives were defined based on the research gaps identified in the literature review:

1. To review at least five recent research papers on academic risk prediction and formally identify the explainability and deployment gaps in existing literature.
2. To define academic risk using Indian grading standards: GPA below 2.0 on a 4.0 scale (equivalent to 50%) = At Risk.
3. To train and compare three machine learning algorithms — Logistic Regression (baseline), Random Forest (existing approach), and XGBoost (proposed) — using consistent preprocessing and evaluation.
4. To prove XGBoost's advantage over existing approaches specifically on recall for at-risk students, which is the most critical metric for early educational intervention.
5. To apply SHAP (SHapley Additive exPlanations) to the trained XGBoost model to generate both global feature importance charts and individual per-student waterfall explanations.
6. To build a Streamlit web application that integrates the trained XGBoost model and SHAP explanations, providing educators with both a risk prediction and its reason for any given student.
7. To evaluate the complete system using accuracy, precision, recall, and F1-score, and document the results with comparison tables and visualizations.

Section 5 | Methodology

The project follows an end-to-end machine learning pipeline consisting of nine stages, each directly connected to the research gaps and objectives defined above.

5.1 Dataset

The Students Performance Dataset published on Kaggle by Rabie El Kharoua was used. It contains 2,392 student records with 15 columns covering demographic, behavioral, and academic features. The dataset had zero missing values, requiring no data cleaning.

Attribute	Detail
Total Records	2,392 students
Total Columns	15 (including GPA and GradeClass)
Missing Values	Zero — completely clean
Feature Types	Demographic, behavioral, family, academic
Target Variable	AcademicRisk (created from GPA)

5.2 Exploratory Data Analysis (EDA)

- GPA distribution visualized with Indian risk threshold (2.0) marked in red
- Risk distribution: 53.26% At Risk (1,274), 46.74% Not At Risk (1,118) — naturally balanced
- Absences showed the strongest negative correlation with GPA: -0.92
- StudyTimeWeekly showed positive correlation with GPA: 0.18
- Scatter plots and correlation heatmap generated for full feature analysis

5.3 Risk Label Creation — Indian Grading

A new binary target column AcademicRisk was created:

- GPA < 2.0 → AcademicRisk = 1 (At Risk) — equivalent to below 50%
- GPA >= 2.0 → AcademicRisk = 0 (Not At Risk) — equivalent to 50% or above

5.4 Preprocessing

- Dropped StudentID (not predictive), GPA (used to define target), GradeClass (causes data leakage)

- All remaining 12 columns were already numeric — no label encoding required
- 80/20 train/test split with random_state=42 for reproducibility
- Training: 1,913 samples | Testing: 479 samples

5.5 Model Training and Comparison

Three models were trained to provide a rigorous comparison:

- Logistic Regression — simple linear baseline
- Random Forest — current standard approach in literature
- XGBoost — proposed approach with hyperparameter tuning via GridSearchCV (5-fold CV)

Best XGBoost parameters found: learning_rate=0.1, max_depth=3, n_estimators=200, subsample=0.8

5.6 Evaluation Results

Model	Accuracy	At Risk Recall	At Risk F1	Role
Logistic Regression	93.95%	0.91	0.92	Baseline
Random Forest	92.69%	0.94	0.93	Existing
XGBoost (Tuned)	**93.32%**	**0.94**	**0.94**	**Proposed** ☑

5.7 SHAP Explainability

- SHAP TreeExplainer applied to tuned XGBoost model
- Global Bar Chart: shows which features matter most across all students
- Bee Swarm Plot: shows how feature values push predictions toward risk or safety
- Waterfall Chart: shows exactly why a specific student was predicted as at-risk

5.8 Model Saving

Tuned XGBoost model saved as academic_risk_xgboost.pkl using joblib for deployment without retraining.

5.9 Streamlit UI

Web app built using Streamlit allows educators to input 12 student features and receive both a risk prediction and a SHAP waterfall explanation for that specific student in real time.

Section 6 | Tools & Technologies

Tool / Library	Version	Purpose
Python	3.10+	Core programming language
pandas	Latest	Data loading, manipulation, and feature engineering
matplotlib	Latest	Chart and plot generation
seaborn	Latest	Statistical visualizations including heatmap
scikit-learn	Latest	Logistic Regression, Random Forest, train_test_split, GridSearchCV, metrics
XGBoost	Latest	Proposed gradient boosting classifier with tuning
SHAP	Latest	Global and per-student explainability via TreeExplainer
joblib	Latest	Saving and loading the trained model (.pkl file)
Streamlit	Latest	Web-based prediction and SHAP explanation interface
VS Code	Latest	Development environment
Kaggle	—	Source of Students Performance Dataset

6.1 Why XGBoost?

- Gradient boosting builds trees sequentially — each tree corrects the errors of the previous
- Built-in L1 and L2 regularization prevents overfitting
- Handles missing values internally
- Consistently outperforms Random Forest on tabular datasets in benchmarks
- Directly compatible with SHAP TreeExplainer for efficient explanation generation

6.2 Why SHAP?

- Mathematically grounded in cooperative game theory (Shapley values)
- Provides both global (population-level) and local (individual-level) explanations
- TreeExplainer is specifically optimized for tree-based models like XGBoost
- Waterfall plots show positive and negative feature contributions per prediction
- More reliable and consistent than alternatives like LIME

Section 7 | Expected Outcomes

7.1 Technical Outcomes

- A trained and tuned XGBoost model achieving 93.32% accuracy and 0.94 recall for at-risk students
- A rigorous comparison table of three ML algorithms with accuracy, precision, recall, and F1-score
- SHAP global feature importance chart identifying Absences and StudyTimeWeekly as top predictors
- SHAP bee swarm plot showing how each feature value influences predictions across all students
- SHAP waterfall chart providing individual per-student explanations for any prediction
- Saved model file (academic_risk_xgboost.pkl) ready for deployment

7.2 Application Outcomes

- A working Streamlit web application deployable in any browser without installation
- Input form accepting 12 student features via sliders and dropdowns
- Real-time prediction output with color-coded risk indicator
- SHAP waterfall explanation showing exactly why the student was predicted as at-risk or safe

7.3 Research Outcomes

- Formal identification and resolution of three research gaps across five reviewed papers
- Proof that XGBoost with tuning outperforms Random Forest on the most critical metric — recall
- Demonstration that SHAP can be embedded in a real-time educator-facing tool
- A complete end-to-end explainable AI system for Indian academic risk prediction

7.4 Evaluation Metrics

Metric	XGBoost Result	Why It Matters
Accuracy	93.32%	Overall correctness of predictions
At Risk Recall	0.94	Catches 94% of all at-risk students — most critical metric
At Risk Precision	0.93	93% of flagged students are genuinely at risk
At Risk F1-Score	0.94	Balance between precision and recall
Overall F1	0.93	Consistent performance across both classes

Section 8 | Applications

8.1 Direct Educational Applications

The primary application of this system is in schools, colleges, and universities across India. The Streamlit web application can be used directly by:

- Teachers — to identify which students in their class are at risk before mid-term or final exams and understand the specific reason for each student's risk
- Academic Counselors — to prioritize counseling sessions for students whose SHAP explanation shows risk driven by personal factors like low parental support
- Class Coordinators — to monitor risk levels across an entire batch and identify common patterns
- Department Heads — to assess overall academic health of a cohort and plan interventions

8.2 Early Warning Systems

The system can serve as the core engine of an Early Warning System (EWS) in educational institutions. By running the prediction model at the start of each semester using available student data, institutions can automatically flag at-risk students and trigger intervention workflows — counseling sessions, parent meetings, tutoring assignments — before academic performance deteriorates.

8.3 Parental Engagement

The SHAP explanation component has a unique application in parental engagement. When a student is flagged as at-risk, the SHAP waterfall chart can be shown to parents to clearly communicate which specific factors — such as high absences or low study time — are contributing to their child's risk. This transforms a vague concern into a specific, actionable conversation.

8.4 Policy Making

At an institutional or government level, the global SHAP feature importance charts can inform education policy. If Absences and StudyTimeWeekly are consistently the top predictors of academic risk across thousands of students, policymakers can prioritize attendance enforcement and study support programs.

8.5 Research Extension

This project can serve as a foundation for future research in explainable educational AI. Possible extensions include:

- Integrating with a live school database for real-time monitoring
- Adding more features like mental health indicators or economic background
- Extending to multi-class risk prediction (Low, Medium, High risk levels)
- Deploying to Streamlit Cloud for remote access by multiple institutions
- Comparing SHAP with LIME and other XAI tools for educational explanation quality

Section 9 | Conclusion and Future Work

9.1 Conclusion

This project makes a genuine and clearly defined contribution to the field of educational data mining. Through a systematic review of five recent papers (2025-2026), three specific research gaps were identified: the black box problem in academic risk prediction, the absence of deployable explainable tools for educators, and the lack of Indian grading standards in existing systems.

All three gaps are addressed by this project. The XGBoost classifier, tuned using GridSearchCV, achieves 93.32% accuracy and — more importantly — 0.94 recall for at-risk students, outperforming both Logistic Regression and Random Forest on the most critical metric for educational intervention. SHAP TreeExplainer is applied to produce both global feature importance and individual per-student explanations, solving the black box problem at its root.

The entire system is deployed as a Streamlit web application, making it the first explainable academic risk prediction tool designed for Indian educational standards that is immediately accessible to non-technical educators. Rather than staying confined to a research paper, this project delivers a working tool that teachers and counselors can use today.

9.2 Key Achievements

- 93.32% accuracy and 0.94 recall for at-risk student detection using XGBoost
- Three algorithms rigorously compared with proven XGBoost advantage on recall and F1
- SHAP explanations for both global feature importance and individual student predictions
- Complete Streamlit web application with prediction and SHAP waterfall chart per student
- Three research gaps formally identified and resolved
- Indian grading standard (GPA < 2.0) applied — first in reviewed literature

9.3 Limitations

- Dataset is from Kaggle — not collected from real Indian schools, which may limit direct generalizability
- UI runs locally — not deployed to a cloud server accessible to multiple institutions simultaneously
- Model does not update automatically when new student data becomes available
- Binary risk classification only — does not distinguish between degrees of risk

9.4 Future Work

- Deploy the application to Streamlit Cloud for remote multi-institution access
- Collect real student data from Indian schools to retrain and validate the model
- Extend to multi-class risk prediction: Low Risk, Moderate Risk, High Risk
- Add LIME alongside SHAP for comparative explainability analysis
- Integrate with school management systems for automated real-time risk monitoring
- Explore deep learning approaches (LSTM, Transformer) for time-series student performance data
- Build a mobile-friendly version of the application for teacher use on smartphones

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